

Functional Safety Analysis of the RaSTA Protocol

Understanding how RaSTA enables safe, reliable, and interoperable communication in railway systems.



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SAFE
COMMUNICATION



RELIABLE
DATA EXCHANGE



INTEROPERABLE
SOLUTIONS



BUILT FOR
RAILWAYS



AGENDA

An overview of the key topics we will cover in this presentation.

1



INTRODUCTION

- Functional Safety
- What is RaSTA?

2



WHY RaSTA?

- Communication Threats
- Why not TCP/IP?

3



RaSTA ARCHITECTURE

- RaSTA Deployment in Railway Systems
- Inside a RaSTA Communication Node

4



HOW RaSTA WORKS

- Connection Establishment
- Safe Data Transmission
- Connection Supervision

5



KEY SAFETY MECHANISMS

- Sequence Numbers
- Integrity Checking
- Heartbeats
- Adaptive Timing
- Retransmission
- Redundancy

6



OPERATIONAL EXAMPLES

- Lost Packet Detection & Retransmission
- Corrupted Message Detection
- Delayed Message Detection
- Heartbeat Failure
- Primary Network Failure

7



REAL-WORLD APPLICATIONS

- Railway Signalling Systems
- ETCS
- Interlockings
- Control Centres

8



ADVANTAGES & LIMITATIONS

- Advantages of RaSTA
- Limitations and Considerations

9



CONCLUSION & DISCUSSION

- Key Takeaways
- Questions & Discussion



“ RaSTA ensures safe, reliable and deterministic communication for the future of railways.”



RaSTA is designed to keep railway communication **safe, reliable** and **available** – even in the presence of faults.

WHAT IS FUNCTIONAL SAFETY?

Ensuring **safety by design**, even when things go wrong.



Functional Safety is part of the overall safety of a system. It refers to a system's ability to **respond to failures** in a safe and controlled way, **reducing risks** to people, the environment, and equipment.



GOAL

Prevent hazardous events or minimize their impact when failures occur.



KEY PRINCIPLES



SAFETY BY DESIGN

Safety is considered in every phase of the system lifecycle, from concept to operation.



IDENTIFY & MANAGE RISKS

Potential hazards are identified and risks are analyzed and reduced to an acceptable level.



DETECT FAILURES

The system can detect faults in time using diagnostics and monitoring.



RESPOND SAFELY

When a fault is detected, the system responds in a safe and controlled manner.



RELIABLE & AVAILABLE

The system continues to operate reliably and is available when needed.

WHY IT MATTERS IN RAILWAYS



Protects passengers, staff, and the public.



Ensures safe operation even in the presence of equipment or communication failures.



Builds trust in railway systems through dependable and predictable behavior.



IN SIMPLE TERMS



Failures can happen (hardware, software, communication, etc.)



Functional Safety ensures these failures are detected in time.



The system reacts in a safe way to prevent accidents or harm.



Result: People, environment and assets are protected.



This is **Functional Safety**
— Safety through reliability.

WHAT IS RaSTA?

RaSTA stands for **Rail Safe Transport Application**.



RaSTA is a safety-oriented communication protocol used in railway applications. It ensures that data is transmitted **reliably and safely** between devices, **even in the presence of communication failures**.



KEY PURPOSE



SAFETY FIRST

Detects communication errors and prevents them from leading to unsafe situations.



RELIABLE COMMUNICATION

Ensures messages are delivered in the correct order, without loss or duplication.



CONTINUOUS SUPERVISION

Monitors the connection constantly using heartbeats and sequence numbers.



SECURE & DETERMINISTIC

Provides integrity protection and predictable behavior for safety-critical systems.



BUILT FOR RAILWAYS

Designed specifically for railway control and signalling applications with high safety demands.

STANDARDIZED BY



RaSTA is standardized in
DIN VDE V 0831-200
(RaSTA Specification)

It defines the protocol requirements to achieve safe and reliable communication in railway systems.



IN SIMPLE TERMS



RaSTA enables devices in a railway system



to communicate safely and reliably



by detecting errors quickly and ensuring correct data exchange



so that railway operations remain safe, dependable and always under control.



RaSTA = Safe Communication for Safer Railways

WHY RaSTA IS NEEDED

RaSTA ensures safe, reliable and secure communication in railway systems.

1 Prevents Hazardous Failures

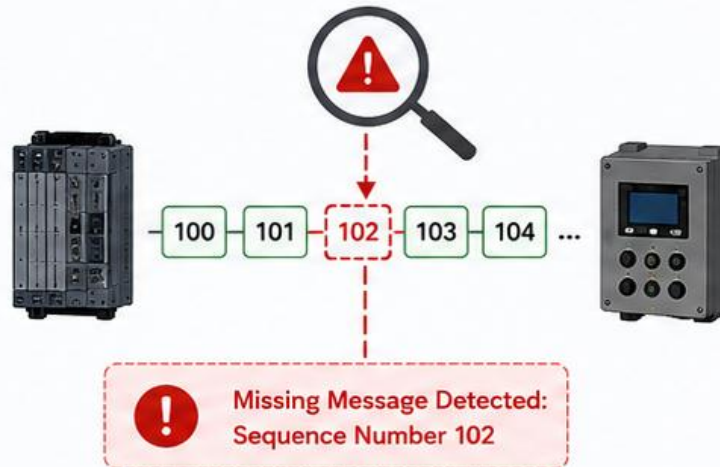
Ensures that communication errors do not lead to accidents or unsafe train operations.



RaSTA keeps communication safe so **failures** never become **accidents**.

2 Detects Faults Quickly

Identifies communication or system errors such as lost, corrupted or out-of-order messages.



RaSTA detects problems **early** before they impact operations.

3 Moves to a Safe State

Stops or disconnects the system when a fault is detected to prevent any unsafe action.



RaSTA ensures the system switches to a **safe state** automatically.



These capabilities make RaSTA essential for **reliable and safe** railway communication.

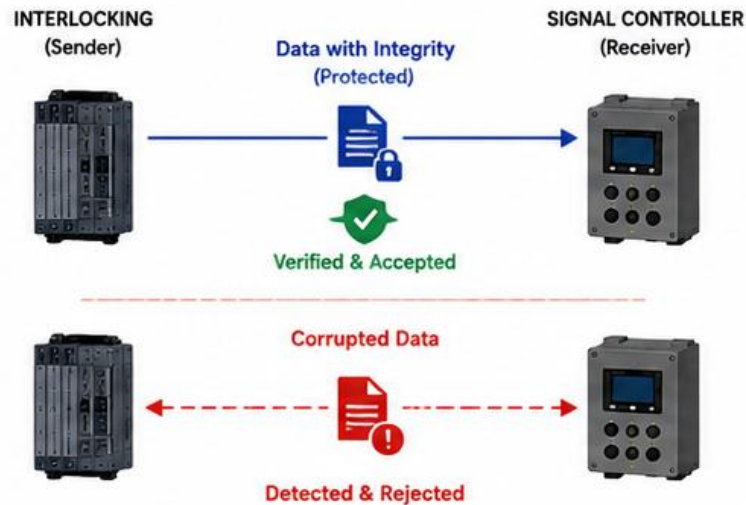


WHY RaSTA IS NEEDED – NEXT 3

RaSTA ensures safe, reliable and secure communication in railway systems.

4 Ensures Data Integrity

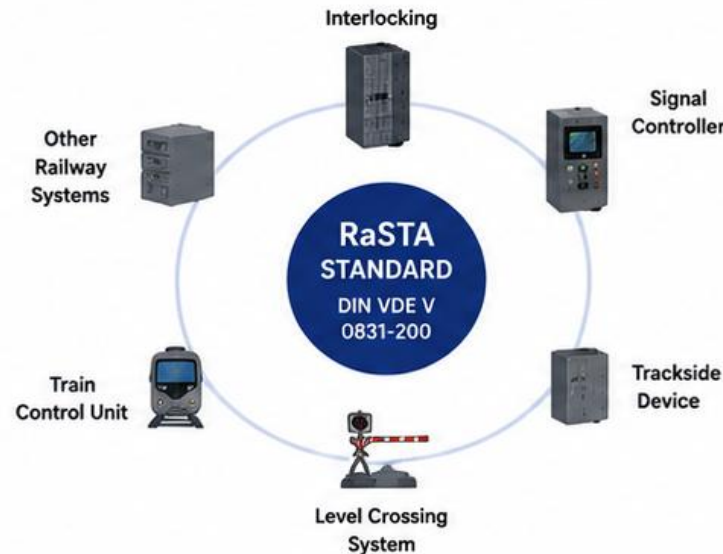
Guarantees that the data received is exactly what was sent, without any alteration or corruption.



RaSTA uses security mechanisms to ensure data is **correct and trustworthy**.

5 Supports Interoperability

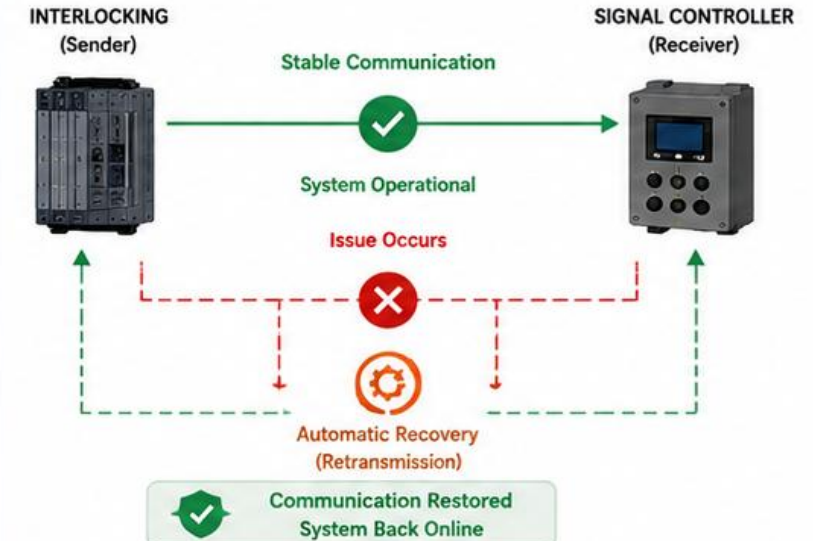
Standardized by DIN VDE V 0831-200, enabling different devices and systems to communicate seamlessly.



RaSTA allows devices from different manufacturers to **work together** reliably.

6 Enhances System Availability

Minimizes downtime by ensuring continuous monitoring and automatic recovery from communication issues.



RaSTA keeps the system **available** and reduces service interruptions.

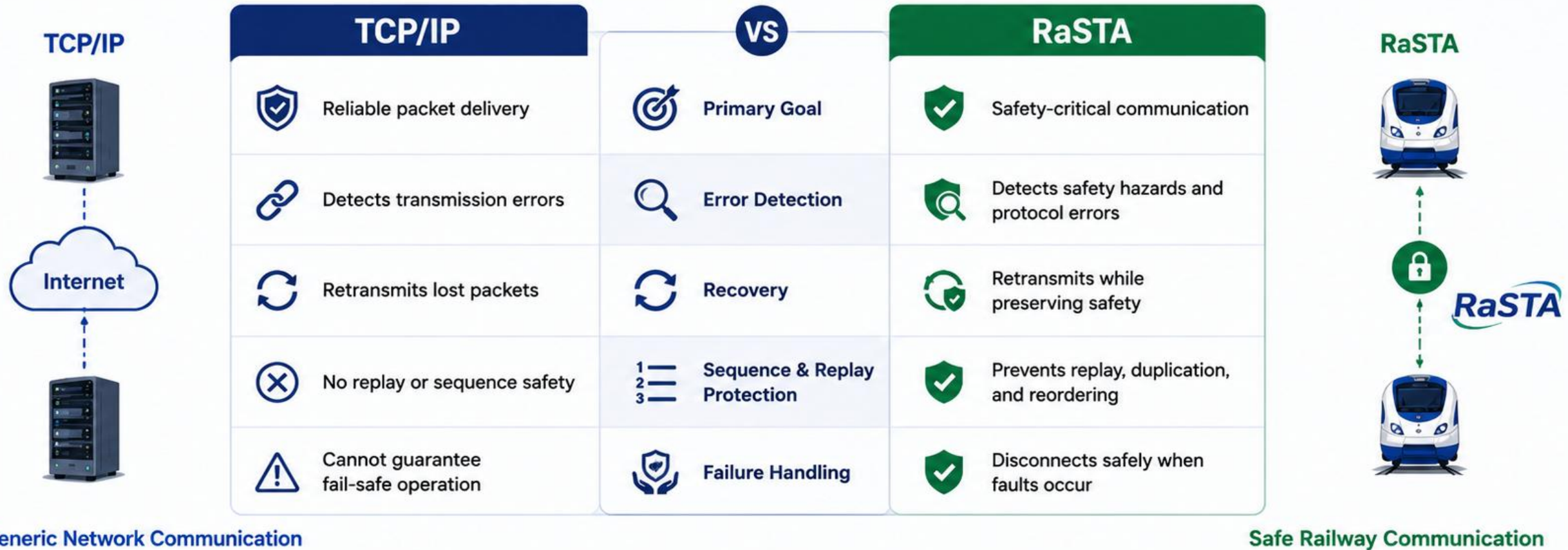


Together, these capabilities make RaSTA essential for **safe, reliable, secure and interoperable** railway communication.



WHY NOT TCP/IP?

TCP provides reliable delivery, but **not functional safety**.



RaSTA builds on standard networks but adds the **safety mechanisms** required for **railway signalling**.



RaSTA ARCHITECTURE

A **safe**, **reliable** and **independent** communication system for railway applications.

WHAT IS RASTA?



RaSTA (Rail Safe Transport Application) is a safety-oriented communication protocol standardized in DIN VDE V 0831-200.

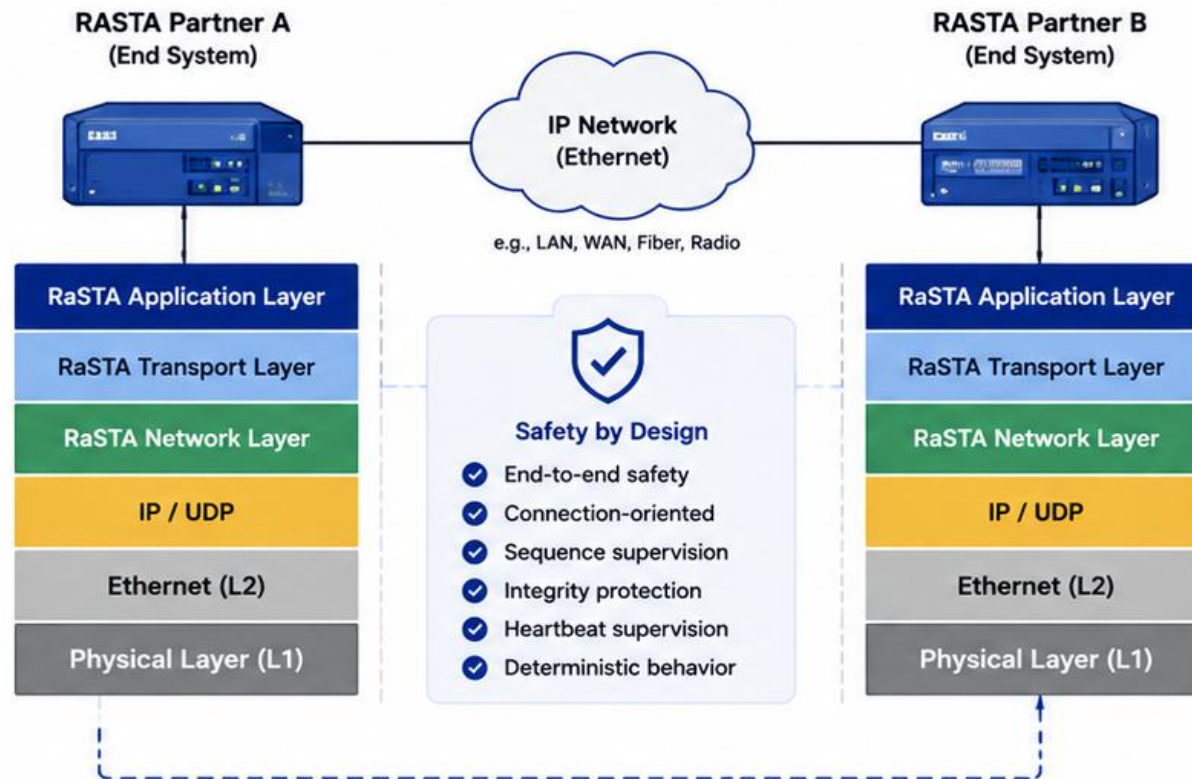


It runs over standard IP networks but adds its own safety mechanisms.



Designed for safety-critical railway functions where communication failures must be detected and handled safely.

RASTA COMMUNICATION ARCHITECTURE



LAYERS & COMPONENTS



RaSTA Application Layer

Provides services to the application. Defines communication endpoints and data handling.



RaSTA Transport Layer

Manages connections, segmentation, acknowledgements and flow control.



RaSTA Network Layer

Handles addressing, routing information and path management.



IP / UDP

Standard IP network and UDP are used as the transport medium.



Ethernet (L2)

Data link layer for frame delivery on the local network.



Physical Layer (L1)

Physical transmission of bits over copper, fiber or wireless media.

KEY CHARACTERISTICS



Safety-focused communication protocol



Independent of underlying IP network



Interoperable between devices from different manufacturers



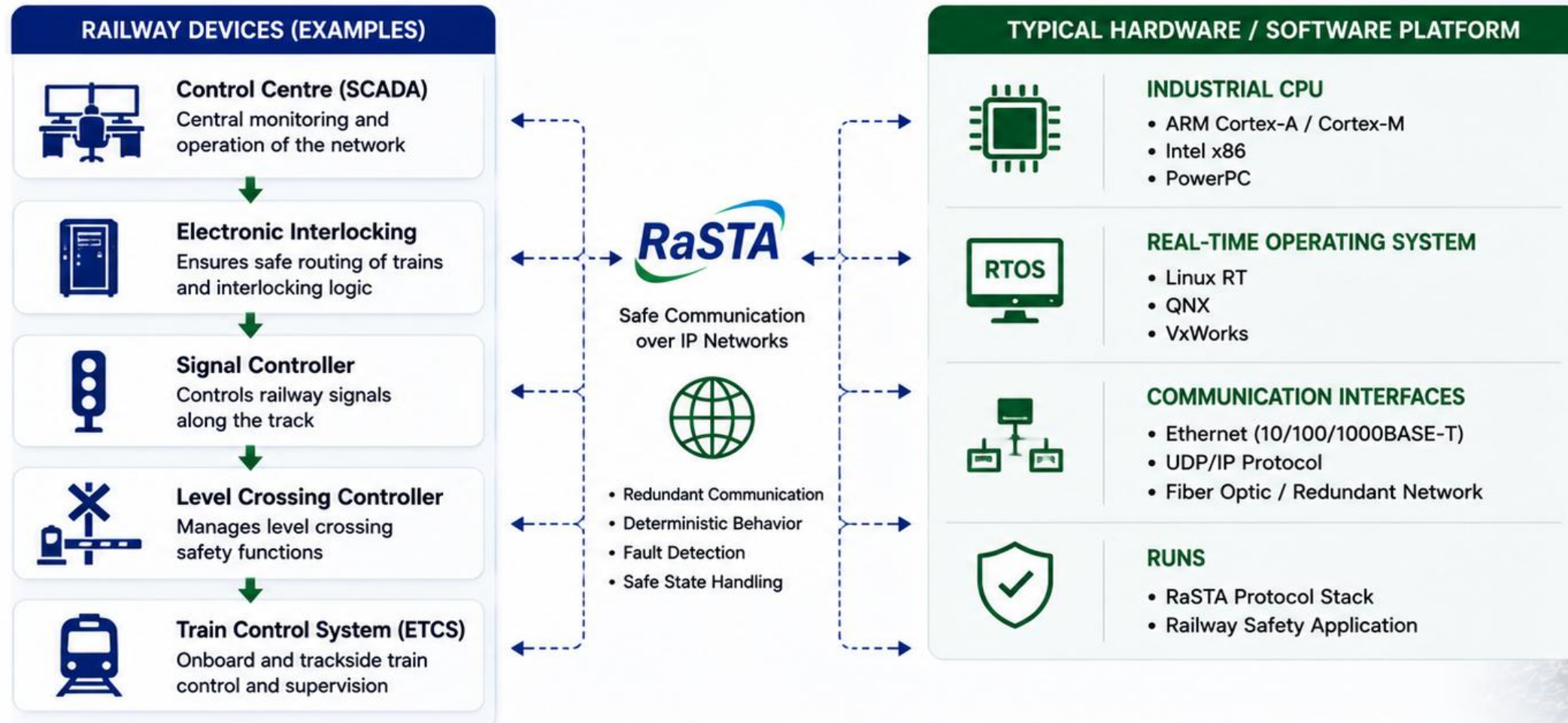
High availability through redundancy and supervision



Meets railway standards and safety requirements

RaSTA DEPLOYMENT IN RAILWAY SYSTEMS

RaSTA enables safe, reliable and interoperable communication between safety-critical railway devices.



RaSTA is a software protocol running on standard industrial hardware platforms. It provides **safe**, **reliable** and **interoperable** communication for mission-critical railway applications.



INSIDE A RaSTA COMMUNICATION NODE

RaSTA is implemented as a set of software modules that work together to provide safe, reliable and deterministic communication.

KEY RESPONSIBILITIES



SAFETY FIRST

Ensures that communication failures do not lead to unsafe behavior.



CONTINUOUS MONITORING

Every message is supervised for correctness, timing and integrity.



FAULT HANDLING

Detects faults, tries safe recovery and switches to safe state if needed.



DETERMINISTIC BEHAVIOR

Guarantees predictable and timely communication.



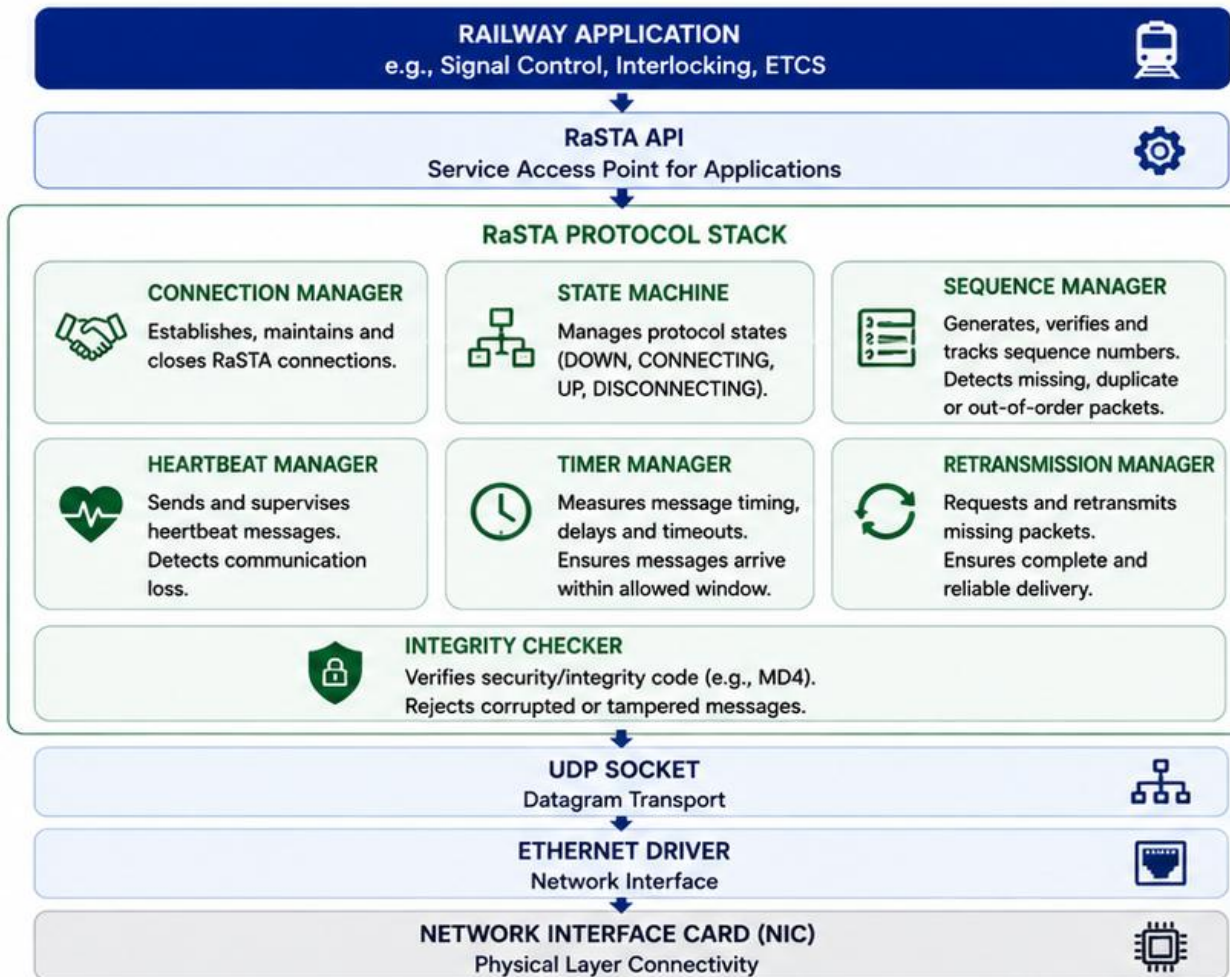
INTEROPERABLE

Works with other RaSTA nodes across standard IP networks.



All modules work together to guarantee functional safety:

Detect faults → Try **safe** recovery → If not possible → Move system to **safe state**.



HOW IT WORKS

- 1 Application creates a message (e.g., set signal to GREEN).
- 2 RaSTA stack adds safety information, checks state and prepares the message.
- 3 Message is sent over UDP/IP through the network.
- 4 Receiver runs the same checks (sequence, timing, integrity).
- 5 If everything is correct, the application receives the data.



TYPICAL HARDWARE PLATFORM

- Industrial CPU: ARM Cortex-A / Intel x86 / PowerPC
- Real-Time OS: Linux RT / QNX / VxWorks
- Interfaces: Ethernet, Fiber Optic, Redundant Network

HOW RaSTA WORKS | FIRST 3 STEPS

Establishing a **safe connection** between two RaSTA communication partners.



RaSTA connections are connection-oriented. Before any data is exchanged, the partners establish a **supervised and safe communication channel**.



1

CONNECTION REQUEST (CR)

Initiate connection



Partner A sends a Connection Request (CR) to Partner B.

Contains:

- Protocol version
- Sender identity
- Supported options
- Initial sequence number
- Integrity protection (MD4)



Purpose: Ask to establish a RaSTA connection and propose communication parameters.

2

CONNECTION RESPONSE (CC)

Accept or reject



Partner B evaluates the request and sends a Connection Response (CC).

Contains:

- Accept / Reject status
- Selected options
- Initial sequence number
- Integrity protection (MD4)



Purpose: Confirm or reject the connection and agree on communication parameters.

3

CONNECTION CONFIRM (CN)

Finalize connection



Partner A confirms the response with a Connection Confirm (CN).

Contains:

- Confirmation of parameters
- Initial sequence number
- Integrity protection (MD4)



Purpose: Finalize the connection. The connection is now established and supervised.



After Step 3, the connection is in the UP (UPPER) state.

Heartbeats are exchanged to supervise the connection and ensure both partners are alive.



HOW RaSTA WORKS | NEXT 3 STEPS

Reliable data exchange, continuous supervision, and safe termination.



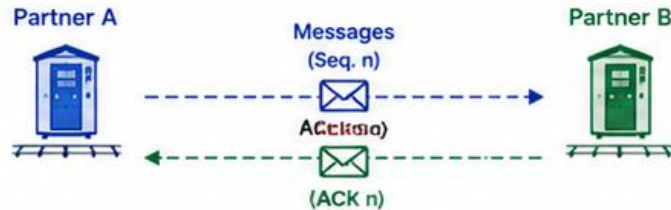
After the connection is established, **data is exchanged safely**.
The connection is **continuously supervised** using heartbeats.
If communication can no longer be guaranteed, the connection is **closed safely**.



4

DATA TRANSFER

Exchange messages safely



Application data is sent with sequence numbers.
The receiver acknowledges each message.

Key Features:

- Sequence number supervision
- Integrity protection (MD4)
- Duplicate and out-of-order detection
- Retransmission if required



Purpose: Ensure reliable and ordered delivery of safety-critical data.

5

HEARTBEAT SUPERVISION

Keep connection alive



Both partners send heartbeats at regular intervals.
If a heartbeat is missed, a fault is assumed.

Key Features:

- Adaptive heartbeat timing
- Missed heartbeat detection
- Triggers safety actions if supervision fails

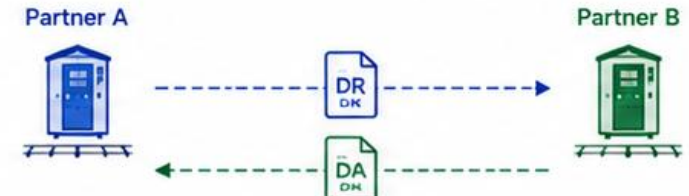


Purpose: Detect communication failures early and maintain connection supervision.

6

DISCONNECTION

Close connection safely



One partner initiates a Disconnect Request (DR).
The other confirms with a Disconnect Acknowledge (DA).

Key Features:

- Controlled connection termination
- Confirmation ensures both sides close safely
- Resources are released



Purpose: Terminate the connection in a controlled and safe manner.



RaSTA ensures that communication is **reliable**, **supervised**, and **safe** from connection establishment to disconnection.



1-3
Connection
Setup



4
Data Transfer



5
Heartbeat
Supervision



6
Disconnection



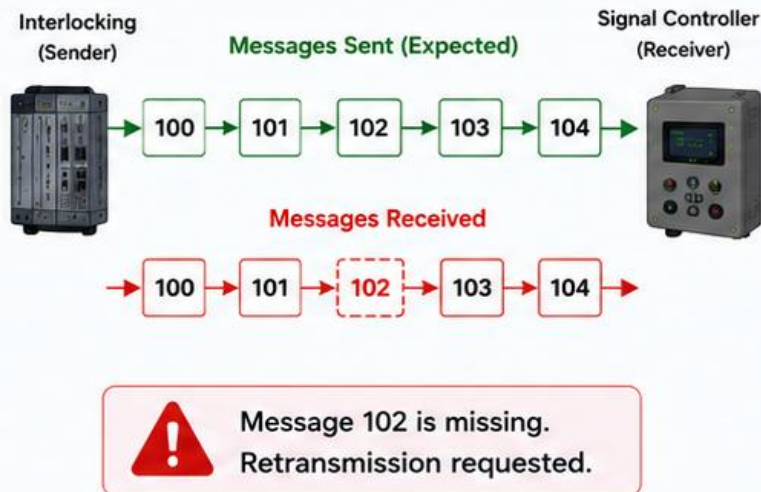
Safe & Reliable
Communication

KEY SAFETY MECHANISMS – FIRST 3

1 SEQUENCE NUMBERS (SN)

Detects lost, duplicate, or out-of-order messages.

Example:



Purpose: Ensures that all messages arrive in the correct order without loss or duplication.

2 INTEGRITY CHECKING (SECURITY CODE)

Detects corrupted or modified messages.

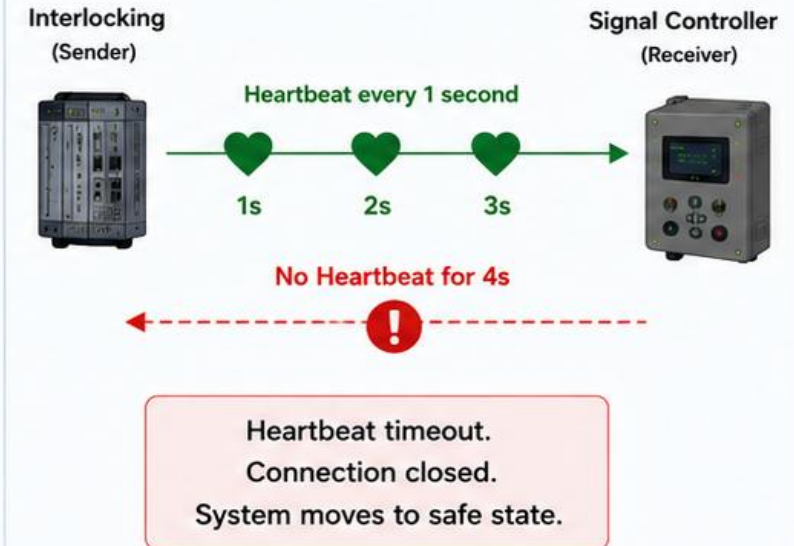
Example:



Purpose: Guarantees message integrity by verifying that data is not corrupted or altered.

3 HEARTBEATS (HB)

Ensures both devices are still communicating.



Purpose: Monitors the connection health and detects failures quickly.



These mechanisms work together to detect faults, ensure reliable communication, and move the railway system to a safe state.



KEY SAFETY MECHANISMS – NEXT 3

4

ADAPTIVE TIMING

Detects messages that arrive too late or too early.

Example:



Allowed Delay
 ≤ 200 ms

Message arrived
within 200 ms
✓ Accepted



Message Arrived After
450 ms

Message discarded
⚠ Considered unsafe



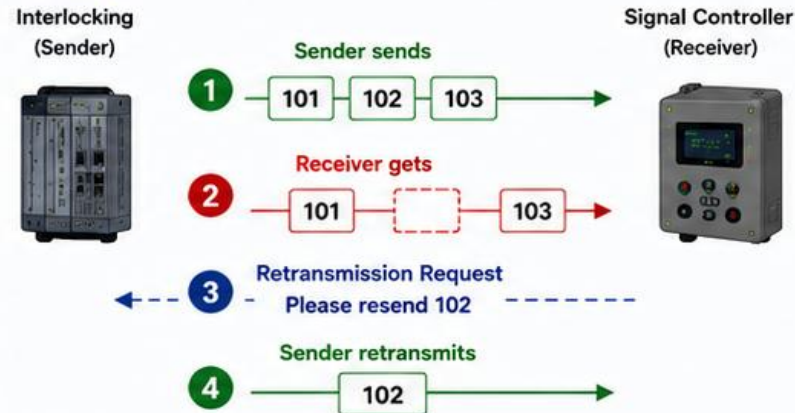
Purpose: Prevents outdated or premature messages from affecting system behavior.

5

RETRANSMISSION

Recovers lost messages by requesting and resending them.

Example:



Result: Missing messages are recovered and the communication remains complete and reliable.



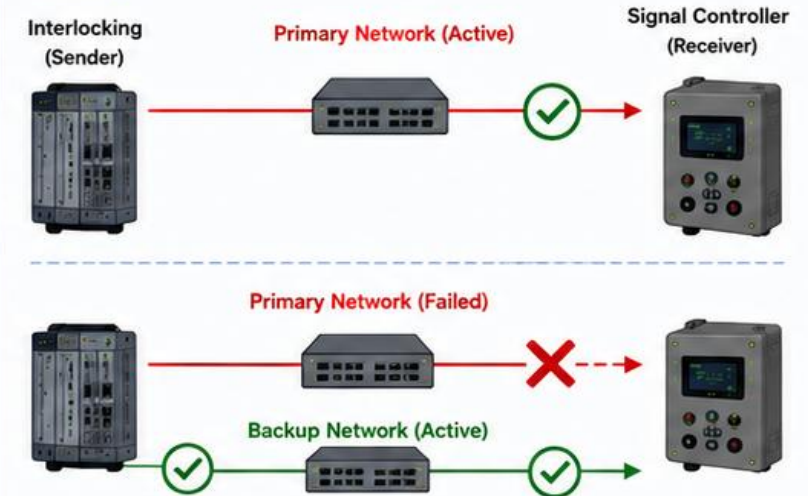
Purpose: Ensures no message is lost and the sequence is complete.

6

REDUNDANCY

Maintains communication if one network channel fails.

Example:



Purpose: Provides continuous communication by switching to a backup channel when the primary channel fails.



These mechanisms ensure reliable, complete, and timely communication — keeping railway operations safe and dependable.



LOST PACKET DETECTION & RETRANSMISSION

RaSTA ensures complete and ordered delivery of safety-critical messages.

SCENARIO



The Interlocking (Sender) sends a command to the Signal Controller (Receiver):

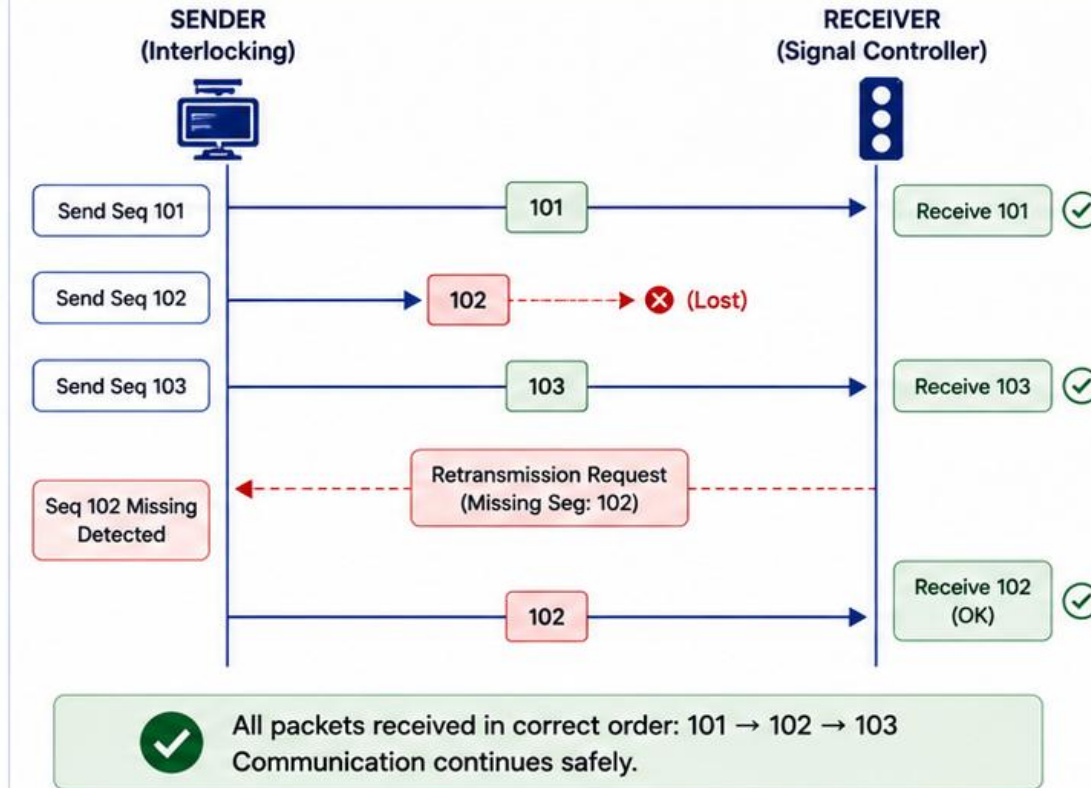
Command: "Set Signal 12 = GREEN"

PACKETS SENT BY SENDER



✓ Delivered ✗ Lost in network

WHAT HAPPENS ON THE NETWORK



PROBLEM

Packet with Sequence Number 102 is lost during transmission. Receiver detects the missing sequence.



HOW RaSTA SOLVES IT

- Sequence Numbers detect missing packets.
- Receiver sends a Retransmission Request.
- Sender retransmits the missing packet.
- Communication continues safely.



TECHNOLOGY USED

- Sequence Numbering
- ACK / Retransmission Mechanism
- Timers (Retransmission Timeout)
- UDP over IP
- Integrity Code (MD4)



IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Incomplete commands executed.
- Wrong signal aspect or missing safety command.
- Possible collisions or derailments.
- Risk to passengers, staff and infrastructure.



KEY TAKEAWAY

RaSTA detects any **lost packet** using sequence supervision and automatically **recovers** it through retransmission. This ensures that safety-critical information is always **complete** and **reliable**.



RaSTA does not ignore failures. It detects them, recovers safely, or stops communication.



CORRUPTED MESSAGE DETECTION

RaSTA ensures the integrity of every message and rejects any data that has been altered.



SCENARIO



The Interlocking sends a speed limit command to the Train Control System (ETCS).

Intended Message:
Speed Limit = 80 km/h

DURING TRANSMISSION

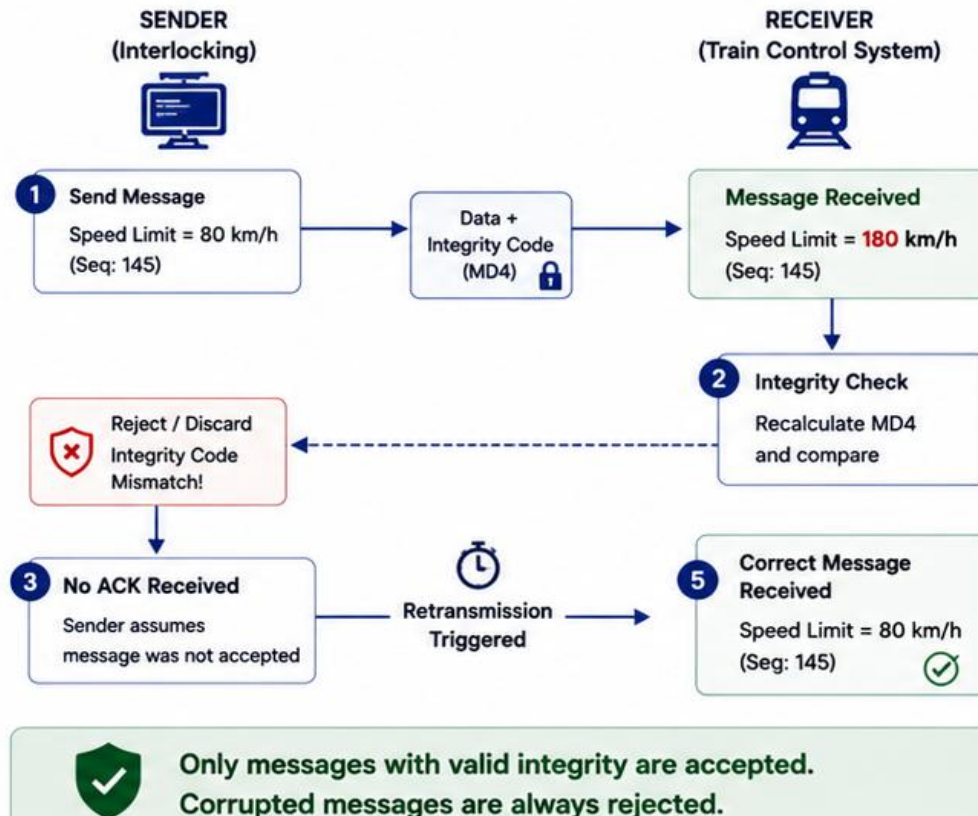
Due to noise or hardware fault, some bits are changed in the data.

Field	Intended Data	Corrupted Data
Speed Limit	80 km/h	180 km/h
Binary (Example)	0101 0000	1011 0100

■ Original / Correct Data

■ Corrupted Data

WHAT HAPPENS ON THE NETWORK



PROBLEM

- Data is corrupted during transmission.
- Receiver cannot trust the content.
- Using wrong data can lead to dangerous decisions.



HOW RaSTA SOLVES IT

- Every message contains an Integrity Code (MD4).
- Receiver recalculates the code.
- If codes don't match, the message is rejected.
- Sender retransmits the message.



TECHNOLOGY USED

- Integrity Code (MD4)
- CRC/Hash based verification
- Sequence Numbers
- ACK / NACK Mechanism
- Timers for Retransmission



IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Train receives wrong speed limit (e.g., 180 instead of 80).
- Overspeed may occur.
- Risk of derailment, collision or infrastructure damage.
- Threat to passenger safety and railway operations.



KEY TAKEAWAY

RaSTA protects message **integrity** using cryptographic verification.
Corrupted data is detected, **rejected**, and retransmitted – ensuring **safe** and **reliable** communication.



*RaSTA does not deliver wrong data.
It delivers correct data –
or no data at all.*





DELAYED MESSAGE DETECTION

RaSTA ensures that time-critical messages arrive within an acceptable time window. Messages that are delayed beyond the limit are rejected to prevent unsafe decisions.



SCENARIO



An emergency braking command is sent from the Control Centre to the Train.

Intended Message:
EMERGENCY BRAKE = TRUE

TIMING REQUIREMENT



Max. allowable delay (T_{max}):
500 ms

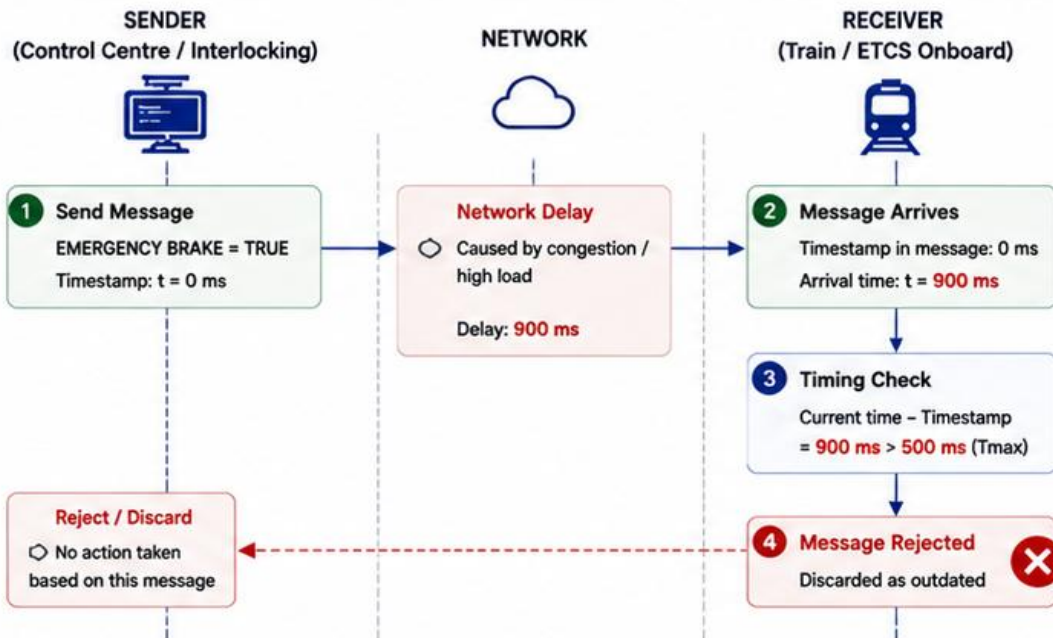
TIMELINE

▶ Message Sent	$t = 0 \text{ ms}$
⌚ Message Arrives	$t = 900 \text{ ms}$
✗ Delay	$900 \text{ ms} (> 500 \text{ ms})$



Message arrives too late!
It is outside the valid time window.

WHAT HAPPENS ON THE NETWORK



Only messages within the allowed time window are accepted.
Late messages are discarded to keep the system safe.



PROBLEM

- Due to network congestion, the message is delayed.
- Late information can no longer represent the real system state.
- Acting on old data can be dangerous.



HOW RaSTA SOLVES IT

- Every message has a Timestamp.
- Receiver checks the age of the message.
- If delay > T_{max} , the message is rejected.
- System continues to operate with safe data.



TECHNOLOGY USED

- Timestamps in each message
- Adaptive Timing (T_{max})
- Timer Manager
- Message Age Verification
- UDP over IP



IF NOT SOLVED - WHAT MIGHT HAPPEN?

- Emergency brake may be applied too late.
- Train may not stop in time.
- Risk of collision, overspeed, or derailment.
- Threat to passenger safety and infrastructure.



KEY TAKEAWAY

RaSTA ensures messages are not only correct, but also **timely**.
Delayed messages are detected and **rejected** to guarantee **safe** and **deterministic** behaviour.



*In safety-critical railway systems,
old information can be as dangerous
as wrong information.*



HEARTBEAT FAILURE

RaSTA continuously supervises the communication link using heartbeat messages. If heartbeats stop, the connection is closed and the system moves to a safe state.



SCENARIO



Two RaSTA nodes are connected between the Interlocking and a Field Device.

Heartbeat Interval:
1 second

WHAT IS A HEARTBEAT?



A small supervision message sent periodically to confirm that the communication partner is alive and the connection is healthy.

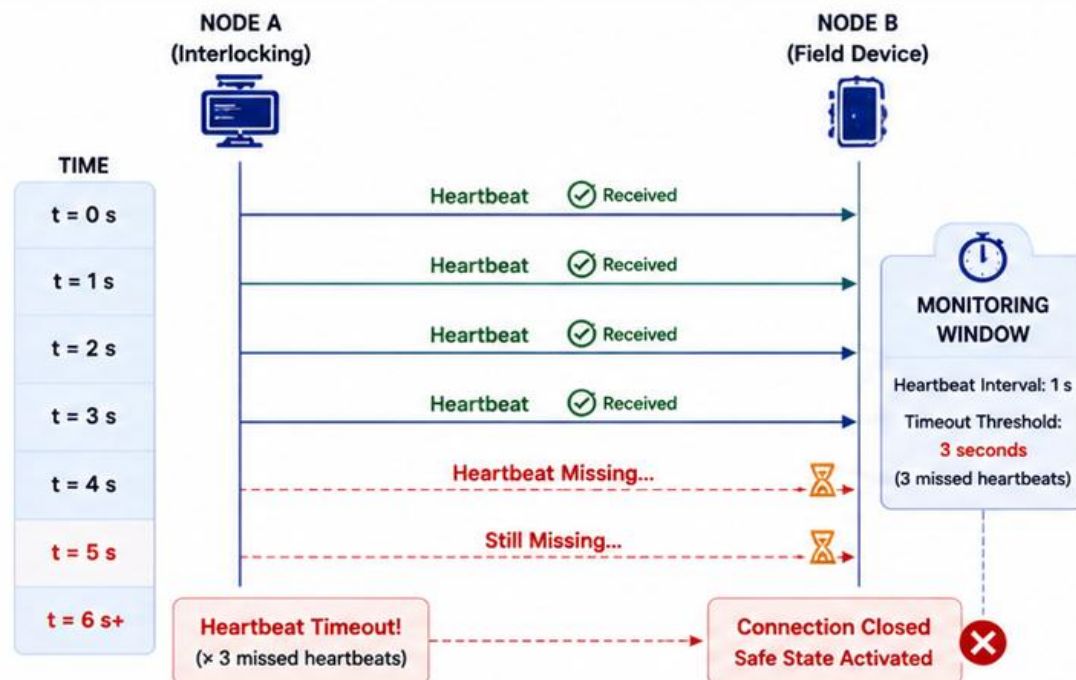
EXPECTED BEHAVIOR

- ✓ Heartbeat received every 1 second
- ✓ Connection remains active

IN THIS SCENARIO

- ✗ Heartbeats stop arriving (for 4 seconds)

WHAT HAPPENS ON THE NETWORK



If heartbeats stop, RaSTA closes the connection and moves the system to a safe state. This prevents uncontrolled or unsafe behavior.



PROBLEM

- Node B stops responding (crash, power loss, or network issue).
- No heartbeat messages are received.
- Connection health becomes unknown.



HOW RaSTA SOLVES IT

- Heartbeat messages are sent periodically.
- If 3 consecutive heartbeats are missed, a timeout occurs.
- Connection is closed automatically.
- System enters a defined safe state.



TECHNOLOGY USED

- Heartbeat Mechanism
- Timer Manager
- Connection State Machine
- Timeout Supervision
- UDP over IP



IF NOT SOLVED – WHAT MIGHT HAPPEN?

- System assumes communication is still alive.
- Old commands may continue to be valid.
- Unsafe movement authority may remain active.
- Risk of collision or derailment.
- Threat to passenger safety and infrastructure.



KEY TAKEAWAY

RaSTA's heartbeat supervision detects communication failures **quickly**. When a failure is detected, the connection is closed and the system moves to a **safe state**, preventing unsafe actions caused by lost or failed communication.

“ No heartbeat = No trust.
No trust = Safe state. ”



PRIMARY NETWORK FAILURE

RaSTA maintains communication continuity by automatically switching to a redundant network when the primary network fails.

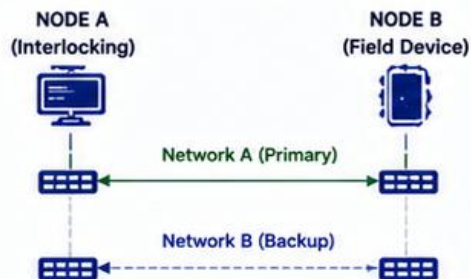
SCENARIO



Two RaSTA nodes communicate using two independent Ethernet networks: Primary and Backup.

During operation, the primary network (Network A) fails.

NETWORK SETUP

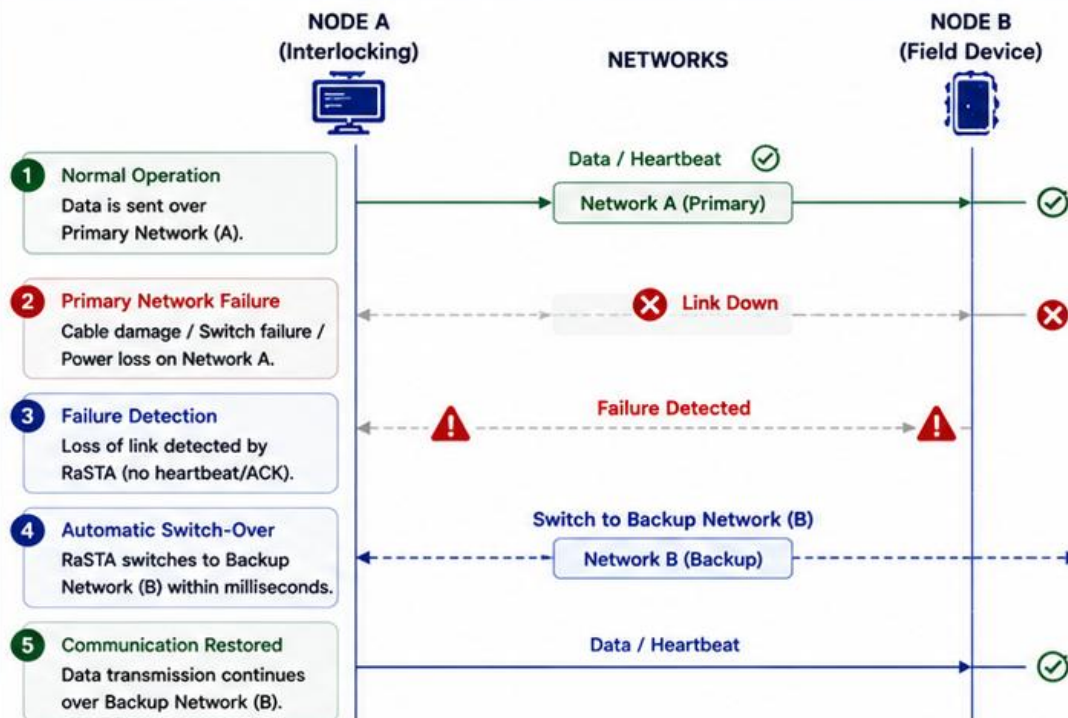


NORMAL OPERATION



All data is transmitted over the Primary Network (A).

WHAT HAPPENS ON THE NETWORK



RaSTA keeps the communication alive by using the redundant path.
No manual intervention is required.



PROBLEM

- The primary network becomes unavailable.
- Communication would be interrupted without redundancy.
- Trains and safety systems could lose critical information.



HOW RaSTA SOLVES IT

- Continuous monitoring of both networks.
- Automatic failure detection.
- Instant switchover to the backup network.
- Communication resumes without data loss (sequence & retransmission handled by RaSTA).



TECHNOLOGY USED

- Redundant Ethernet Networks
- Link Monitoring
- Automatic Channel Switch-Over
- RaSTA Connection State Machine
- Sequence Numbers & Retransmission



IF NOT SOLVED – WHAT MIGHT HAPPEN?

- Loss of communication between critical devices.
- Interlocking may not receive field status.
- Wrong signal aspect or point position.
- Possible collisions, derailments, or service halt.



KEY TAKEAWAY

RaSTA's redundancy mechanism ensures **high availability** and **fault tolerance**.

If the primary network fails, RaSTA automatically uses the backup network to keep railway operations safe.



Redundancy is not an option in railways – it is a requirement for safety and reliability.



REAL-WORLD APPLICATIONS OF RaSTA

RaSTA ensures **safe** and **reliable** communication across modern railway systems.

RaSTA
Safe Communication
for Railways



ELECTRONIC INTERLOCKINGS



- Controls signals, points and track routes
- Exchanges safety-critical commands and status
- Ensures safe route setting



ETCS / TRAIN CONTROL SYSTEMS



- Transmits movement authority and train data
- Provides continuous supervision
- Supports safe train separation



LEVEL CROSSINGS



- Coordinates barriers, lights and warning systems
- Exchanges real-time status information
- Prevents dangerous situations



CONTROL CENTRES



- Connects field devices with operations
- Centralized monitoring and diagnostics
- Enables faster and safer decision making



OTHER RAILWAY APPLICATIONS



- Trackside equipment communication
- Station systems
- Diagnostics and maintenance systems



RaSTA is used wherever **safety-critical** information must be exchanged **reliably**, **deterministically** and **securely**.



ADVANTAGES & LIMITATIONS OF RaSTA

RaSTA is designed to ensure **functional safety** in railway communication.



ADVANTAGES



FUNCTIONAL SAFETY COMPLIANT

Designed according to DIN VDE V 0831-200 and DIN EN 50159 for safety-related communication.



HIGH COMMUNICATION SAFETY

Detects communication faults such as loss, delay, duplication, corruption and replay.



FAIL-SAFE OPERATION

If communication can no longer be trusted, RaSTA disconnects and supports a safe state.



HIGH AVAILABILITY

Redundancy layer allows communication to continue even if one channel fails.



INTEROPERABILITY

Standardized protocol allows equipment from different manufacturers to communicate.



DETERMINISTIC COMMUNICATION

Adaptive timing and supervision provide predictable and timely message delivery.



LIMITATIONS



MORE COMPLEX THAN TCP/IP

RaSTA adds multiple safety mechanisms, making the implementation more complex.



HIGHER COMMUNICATION OVERHEAD

Additional headers, sequence numbers and safety checks increase bandwidth usage.



REQUIRES SYNCHRONIZED IMPLEMENTATION

Correct operation depends on precise implementation and synchronization of both partners.



SPECIFIC TO RAILWAY APPLICATIONS

RaSTA is optimized for railway signalling and may not be suitable for general-purpose systems.



DOES NOT REPLACE APPLICATION-LEVEL SAFETY

Safety also depends on correct application logic and system design.



RaSTA prioritizes **safety** and **reliability** over pure performance, making it ideal for mission-critical railway systems.



SAFE



RELIABLE



INTEROPERABLE



RAILWAY-FOCUSED



MTBF Mean Time Between Failures

Definition

Average operating time before a component fails.

Examples



Industrial Computer



Ethernet Switch



Signal Controller



Fibre Optic Network

Typical Values

CPU	100,000+ hours
Ethernet Switch	150,000+ hours
Fibre Network	200,000+ hours



MTTR Mean Time To Repair

Definition

Average time required to restore operation after failure.

Examples

Automatic Recovery

- Software Restart
- Automatic Reconnection

Manual Recovery

- Replace Switch
- Replace Controller
- Replace Fibre Cable

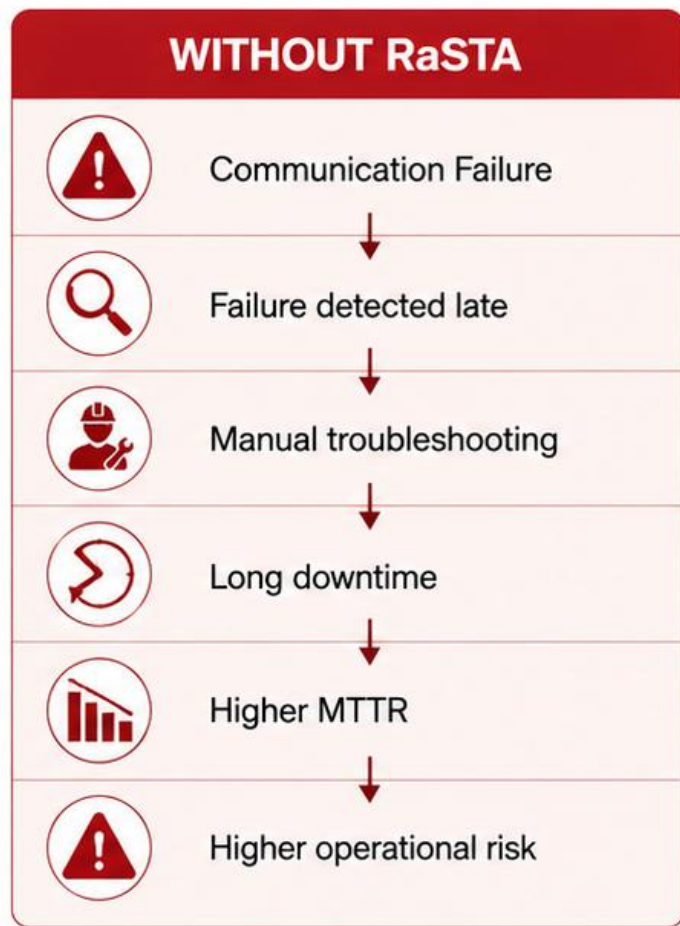
Typical Values

Automatic	Milliseconds to Seconds
Hardware	30 Minutes to Several Hours



MTBF and MTTR describe the reliability of railway hardware and systems, not the RaSTA protocol itself.

How RaSTA Improves System Availability



Comparison		
Aspect	Without RaSTA	With RaSTA
 Fault Detection	Manual fault detection	Automatic fault detection
 Recovery Method	Manual recovery	Automatic recovery
 Communication Outage	Long communication outage	Short communication interruption
 Operational Risk	High	Low
 System Availability	Lower system availability	Higher system availability



RaSTA does not define MTBF or MTTR, but it significantly improves the overall availability and reliability of the railway system by reducing recovery time and detecting faults immediately.

CONCLUSION

RaSTA provides **safe**, **reliable** and **deterministic** communication for safety-critical railway applications.



FUNCTIONAL SAFETY

RaSTA detects communication faults and prevents them from becoming safety hazards.



RELIABILITY & AVAILABILITY

Through sequence monitoring, adaptive timing, retransmission and redundancy, RaSTA ensures high availability.



SECURE & DETERMINISTIC

Integrity protection, replay prevention and deterministic behaviour guarantee trustworthy message exchange.



STANDARDIZED & INTEROPERABLE

Based on DIN VDE V 0831-200 and DIN EN 50159, RaSTA enables interoperability between devices from different manufacturers.



RAILWAY FOCUSED

Designed specifically for railway signalling and control systems, where safety is more important than performance.



RaSTA does not prevent every communication failure— it ensures that communication failures **never** become safety hazards.



Thank you for your attention!
Questions and Discussion

